



Chapter 6: Geometrical Figures

Three-week revision booklet · 6 worksheets · 108 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 1 OF 13

How to use this booklet

There are six worksheets, two for each week, and each one runs across **two pages**. Most questions come with a clear, labelled diagram, so you can see the shape you are working with. Each sheet has **18 questions** split into **Easy, Medium** and **Challenging**, and every section opens with a short hint box telling you exactly how to tackle that kind of question. The Challenging questions are written in words only — read them carefully and **draw your own diagram** in the space provided before you answer. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Classifying triangles
1	2	Angle sum of a triangle
2	3	Exterior angles
2	4	Isosceles triangles and parallel lines
3	5	Quadrilaterals
3	6	Mixed revision

Progress tracker

Worksheet	Date completed	Score / 18	Checked by
1 Classifying triangles			
2 Angle sum of a triangle			
3 Exterior angles			
4 Isosceles triangles and parallel lines			
5 Quadrilaterals			
6 Mixed revision			

Worksheet 1: Classifying triangles

Week 1 · Worksheet 1 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 2 OF 13

By sides: equilateral (3 equal), isosceles (2 equal), scalene (none equal). By angles: acute (all less than 90°), right (one is 90°), obtuse (one more than 90°).

EASY

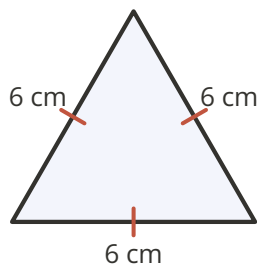
9 QUESTIONS

HOW TO CLASSIFY A TRIANGLE

- Sides with the same number of red tick marks are equal in length.
- 3 equal sides = **equilateral** · 2 equal = **isosceles** · none equal = **scalene**.
- For angles, look only at the **biggest** one: under 90° = acute, exactly 90° = right, over 90° = obtuse.

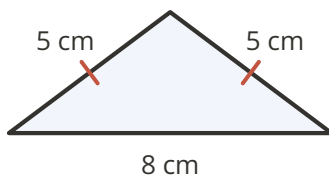
Classify each triangle by its sides.

1. Name this triangle.



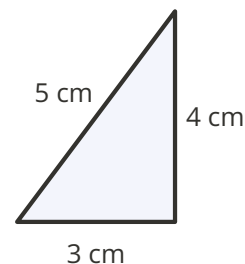
Answer: _____

2. Name this triangle.



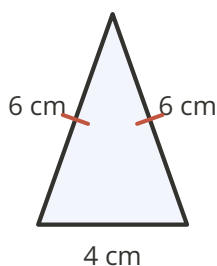
Answer: _____

3. Name this triangle.



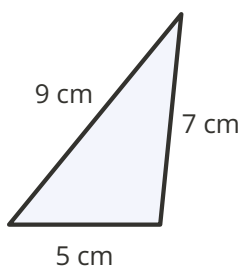
Answer: _____

4. Name this triangle.



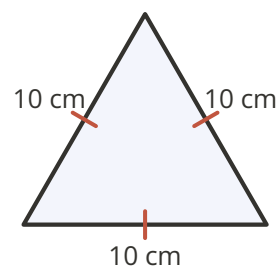
Answer: _____

5. Name this triangle.



Answer: _____

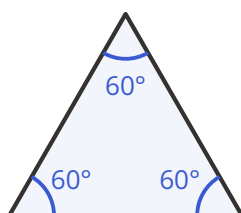
6. Name this triangle.



Answer: _____

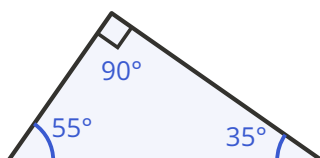
Classify each triangle by its angles.

7. Acute, right or obtuse?



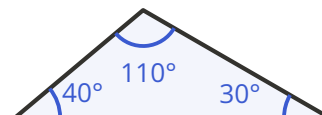
Answer: _____

8. Acute, right or obtuse?



Answer: _____

9. Acute, right or obtuse?



Answer: _____

Worksheet 1: Classifying triangles

Week 1 · Worksheet 1 of 6 · continued

PAGE 3 OF 13

MEDIUM

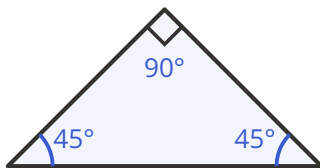
6 QUESTIONS

NAMING A TRIANGLE TWO WAYS

- Equal angles sit opposite equal sides, so two equal angles means it is isosceles.
- Give the side name first, then the angle name — for example “isosceles and obtuse”.

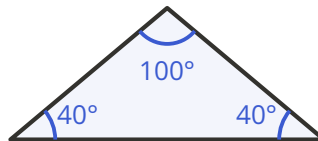
Name each triangle by its sides and by its angles.

10. Name it two ways.



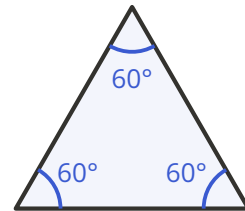
Answer: _____

11. Name it two ways.



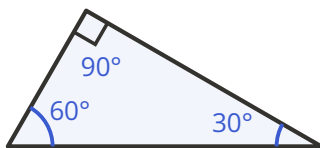
Answer: _____

12. Name it two ways.



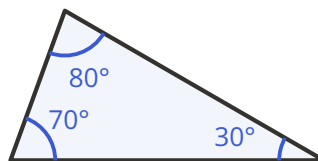
Answer: _____

13. Name it two ways.



Answer: _____

14. Name it two ways.



Answer: _____

15. Name it two ways.



Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Sketch the triangle first, then mark equal sides with ticks and write the angles on.
- A rough sketch is fine — it just has to show the right kind of triangle.

Read each one, draw your own diagram, then answer the question.

16. Draw a triangle that has two equal sides and one obtuse angle. Mark the equal sides, then name the triangle two ways.

Draw your diagram here

Answer: _____

17. A triangle has sides 7 cm, 7 cm and 7 cm. Draw and label it, then name it by its sides and by its angles.

Draw your diagram here

Answer: _____

18. Can a triangle have two right angles? Draw a sketch to show what happens, and explain your answer.

Draw your diagram here

Answer: _____

Worksheet 2: Angle sum of a triangle

Week 1 · Worksheet 2 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 4 OF 13

The three angles of any triangle add to 180° .

EASY

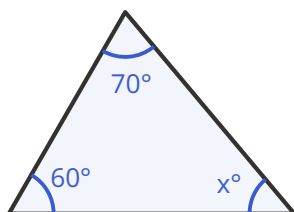
9 QUESTIONS

HOW TO FIND THE MISSING ANGLE

- Add the two angles you are given.
- Take that total away from 180° . What is left is the missing angle.
- Example: $70^\circ + 60^\circ = 130^\circ$, so $x = 180^\circ - 130^\circ = 50^\circ$.

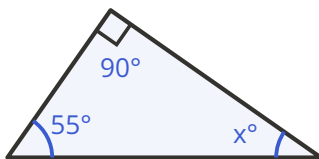
Find the size of the angle marked x.

1. Find x.



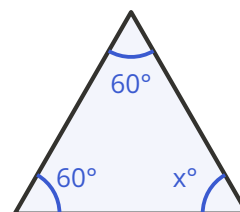
Answer: _____

2. Find x.



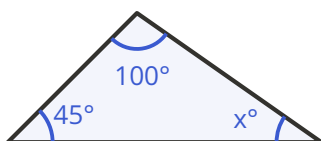
Answer: _____

3. Find x.



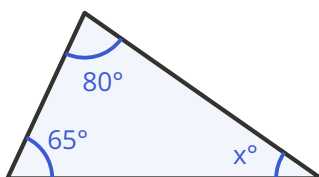
Answer: _____

4. Find x.



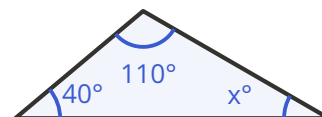
Answer: _____

5. Find x.



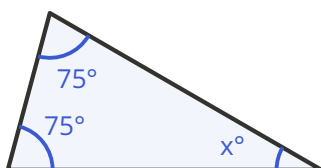
Answer: _____

6. Find x.



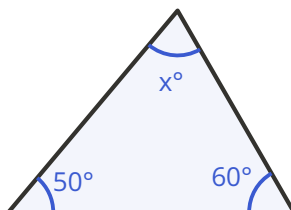
Answer: _____

7. Find x.



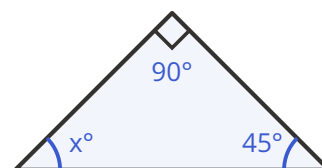
Answer: _____

8. Find x.



Answer: _____

9. Find x.



Answer: _____

Worksheet 2: Angle sum of a triangle

Week 1 · Worksheet 2 of 6 · continued

PAGE 5 OF 13

MEDIUM

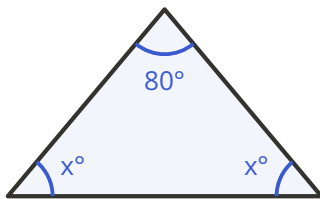
6 QUESTIONS

WHEN THE ANGLES USE X

- Write an equation: add all three angle expressions and make them equal 180.
- Example: $x + 2x + 3x = 180$, so $6x = 180$ and $x = 30$.

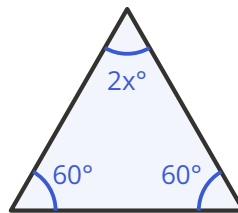
Write an equation and find the value of x .

10. Find x .



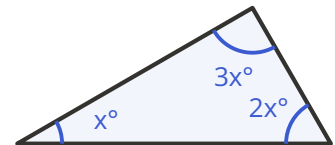
Answer: _____

11. Find x .



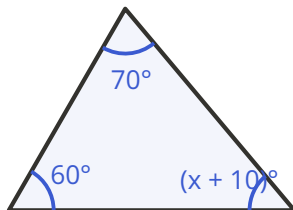
Answer: _____

12. Find x .



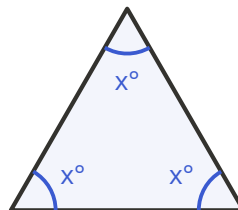
Answer: _____

13. Find x .



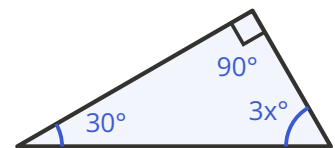
Answer: _____

14. Find x .



Answer: _____

15. Find x .



Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Sketch the triangle and write on every angle you know.
- A right angle is 90° ; in an isosceles triangle the two base angles are equal.

Read each one, draw your own diagram, then answer the question.

16. A right-angled triangle has one angle of 37° . Draw it, mark the right angle, and find the third angle.

Draw your diagram here

Answer: _____

17. The angles of a triangle are in the ratio $1 : 2 : 3$. Draw a sketch and find all three angles.

Draw your diagram here

Answer: _____

18. Explain why a triangle cannot have two obtuse angles. Use a sketch to help your explanation.

Draw your diagram here

Answer: _____

Worksheet 3: Exterior angles

NAME _____

Week 2 · Worksheet 3 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

PAGE 6 OF 13

An exterior angle of a triangle equals the sum of the two opposite interior angles. It is also supplementary to the interior angle beside it.

EASY

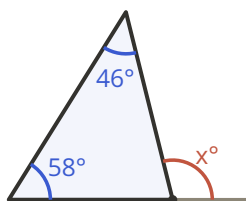
9 QUESTIONS

HOW TO FIND AN EXTERIOR ANGLE

- The exterior angle is made by extending one side (the grey arrow).
- Just **add the two interior angles furthest away** from it.
- Example: $58^\circ + 46^\circ = 104^\circ$, so $x = 104^\circ$.

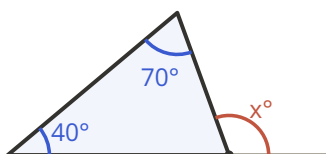
Find the exterior angle marked x.

1. Find x.



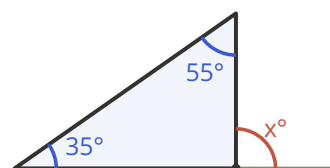
Answer: _____

2. Find x.



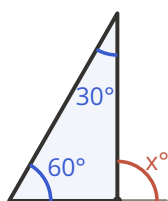
Answer: _____

3. Find x.



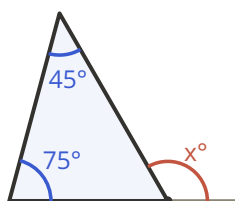
Answer: _____

4. Find x.



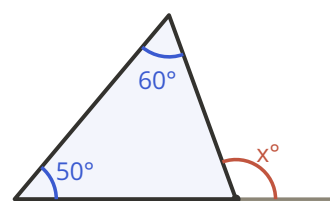
Answer: _____

5. Find x.



Answer: _____

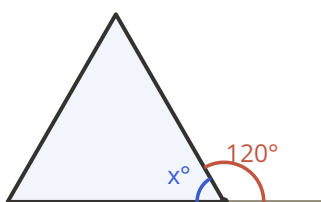
6. Find x.



Answer: _____

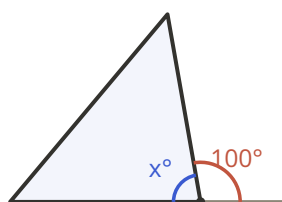
Find the interior angle marked x, next to the exterior angle.

7. Find x.



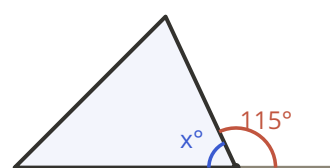
Answer: _____

8. Find x.



Answer: _____

9. Find x.



Answer: _____

Worksheet 3: Exterior angles

Week 2 · Worksheet 3 of 6 · continued

PAGE 7 OF 13

● MEDIUM

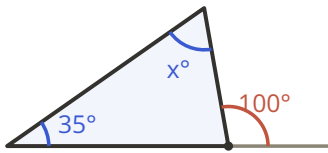
6 QUESTIONS

WORKING BACKWARDS

- If you know the exterior angle, subtract the interior angle you know: $100^\circ - 35^\circ = 65^\circ$.
- The exterior angle and the interior angle next to it always add to **180°**.

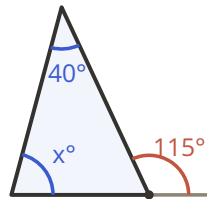
Find the value of x .

10. Find x .



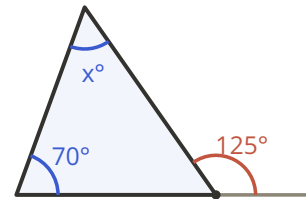
Answer: _____

11. Find x .



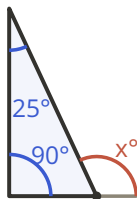
Answer: _____

12. Find x .



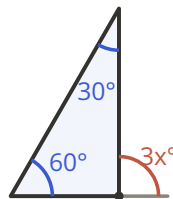
Answer: _____

13. Find x .



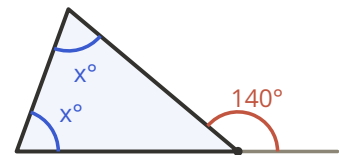
Answer: _____

14. Find x .



Answer: _____

15. Find x .



Answer: _____

● CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw the triangle, then extend one side with a ruler to make the exterior angle.
- Mark the two opposite interior angles — those are the ones you add.

Read each one, draw your own diagram, then answer the question.

16. An exterior angle of a triangle is 130° , and one opposite interior angle is 55° . Draw the triangle and find the other interior angle.

Draw your diagram here

Answer: _____

17. Explain why an exterior angle is always bigger than either of the two opposite interior angles. Use a sketch.

Draw your diagram here

Answer: _____

18. A right-angled triangle has an exterior angle of 145° . Draw it and find all three interior angles.

Draw your diagram here

Answer: _____

Worksheet 4: Isosceles triangles and parallel lines

Week 2 · Worksheet 4 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 8 OF 13

In an isosceles triangle the two angles opposite the equal sides are equal. Parallel lines are marked with matching arrows.

EASY

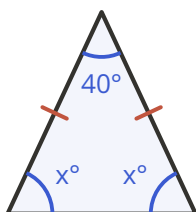
9 QUESTIONS

HOW TO USE AN ISOSCELES TRIANGLE

- The two ticked sides are equal, so the two **base angles** are equal.
- Given the apex: base angle = $(180^\circ - \text{apex}) \div 2$.
- Given a base angle: apex = $180^\circ - 2 \times \text{base angle}$.

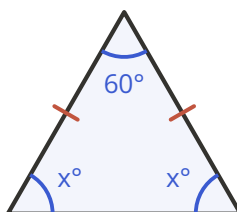
The apex angle is given. Find each base angle x.

1. Find x.



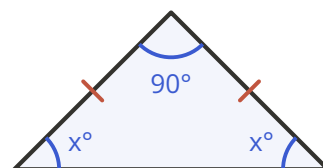
Answer: _____

2. Find x.



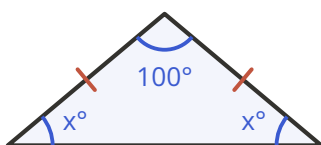
Answer: _____

3. Find x.



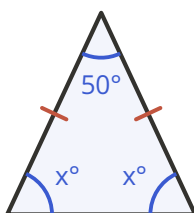
Answer: _____

4. Find x.



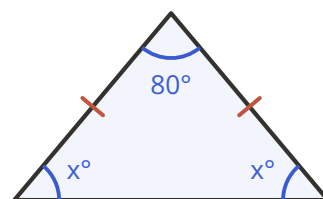
Answer: _____

5. Find x.



Answer: _____

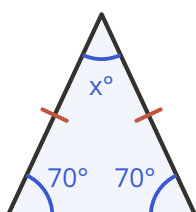
6. Find x.



Answer: _____

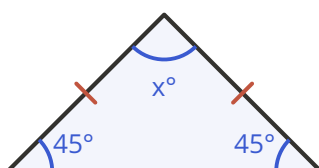
A base angle is given. Find the apex angle x.

7. Find x.



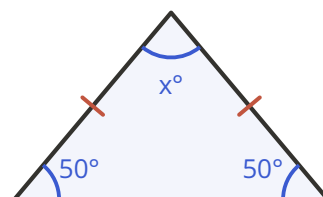
Answer: _____

8. Find x.



Answer: _____

9. Find x.



Answer: _____

Worksheet 4: Isosceles triangles and parallel lines

Week 2 · Worksheet 4 of 6 · continued

PAGE 9 OF 13

MEDIUM

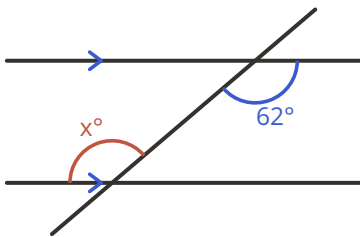
6 QUESTIONS

ANGLES IN PARALLEL LINES

- Z shape: alternate angles are **equal**.
- F shape: corresponding angles are **equal**.
- C or U shape: co-interior angles **add to 180°**.

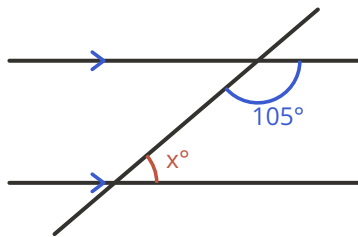
The two lines with arrows are parallel. Find x .

10. Alternate angles. Find x .



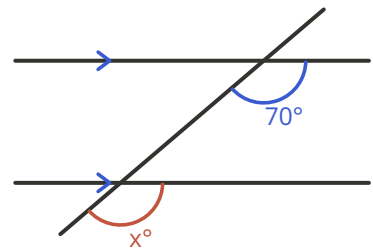
Answer: _____

11. Co-interior angles. Find x .



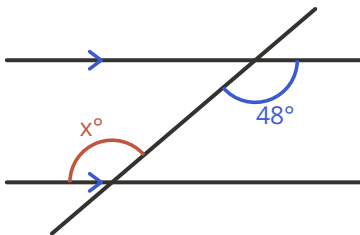
Answer: _____

12. Corresponding angles. Find x .



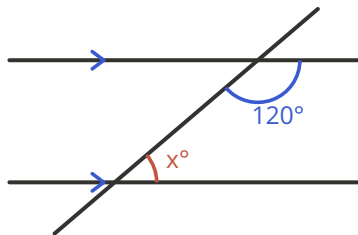
Answer: _____

13. Alternate angles. Find x .



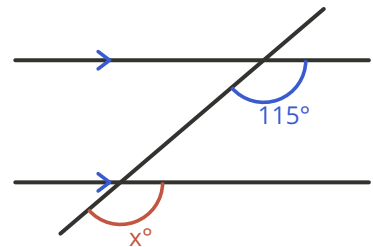
Answer: _____

14. Co-interior angles. Find x .



Answer: _____

15. Corresponding angles. Find x .



Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Mark the equal sides with ticks first — that shows you which angles are equal.
- For parallel-line questions, look for the Z, F or C shape in your drawing.

Read each one, draw your own diagram, then answer the question.

16. An isosceles triangle has an apex angle of 40° . Draw it, mark the equal sides, and find the base angles. Give a reason.

Draw your diagram here

Answer: _____

17. Draw two parallel lines cut by a straight line. Mark a pair of alternate angles and explain why they are equal.

Draw your diagram here

Answer: _____

18. An isosceles triangle has a perimeter of 32 cm and two equal sides of 12 cm. Draw and label it, then find the length of the base.

Draw your diagram here

Answer: _____

Worksheet 5: Quadrilaterals

Week 3 · Worksheet 5 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 10 OF 13

The four angles of any quadrilateral add to 360° . Matching ticks show equal sides and matching arrows show parallel sides.

EASY

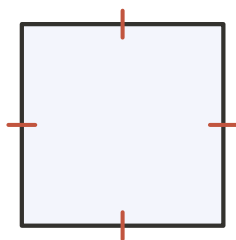
9 QUESTIONS

HOW TO NAME A QUADRILATERAL

- Matching **ticks** = equal sides. Matching **arrows** = parallel sides.
- Square: 4 equal sides + 4 right angles · Rhombus: 4 equal sides, no right angles.
- Rectangle: 4 right angles · Trapezium: exactly one pair of parallel sides.

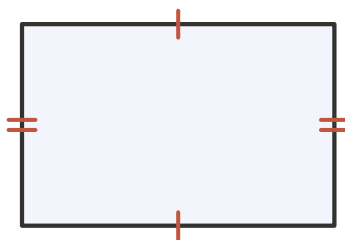
Name each quadrilateral.

1. Name this shape.



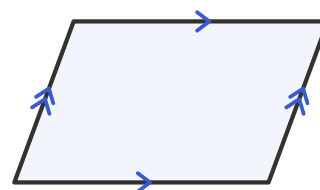
Answer: _____

2. Name this shape.



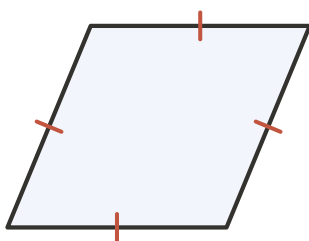
Answer: _____

3. Name this shape.



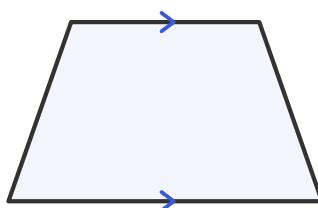
Answer: _____

4. Name this shape.



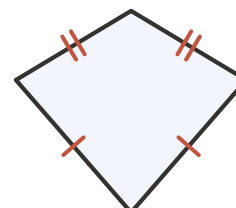
Answer: _____

5. Name this shape.



Answer: _____

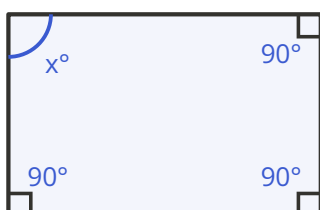
6. Name this shape.



Answer: _____

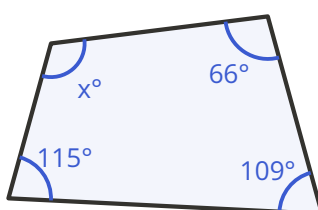
Find the fourth angle x.

7. Find x.



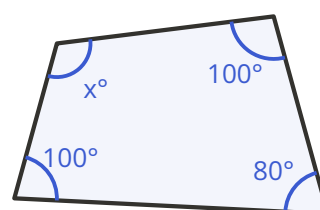
Answer: _____

8. Find x.



Answer: _____

9. Find x.



Answer: _____

Worksheet 5: Quadrilaterals

Week 3 · Worksheet 5 of 6 · continued

PAGE 11 OF 13

MEDIUM

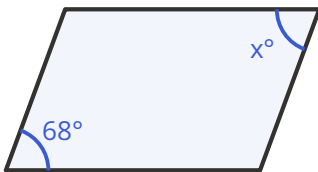
6 QUESTIONS

FINDING A MISSING ANGLE

- All four angles add to 360° , so add what you know and subtract from 360° .
- In a parallelogram, **opposite angles are equal** and angles next to each other add to 180° .

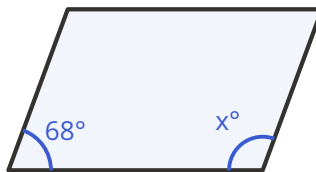
Use the properties of the shape to find x .

10. Opposite angles. Find x .



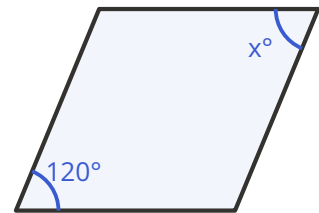
Answer: _____

11. Co-interior angles. Find x .



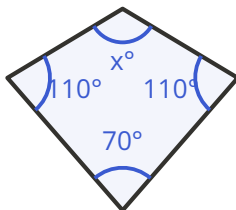
Answer: _____

12. Opposite angles. Find x .



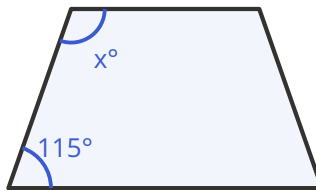
Answer: _____

13. Angle sum. Find x .



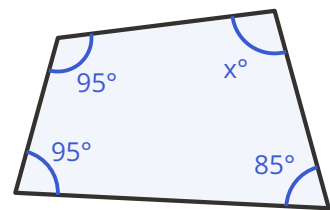
Answer: _____

14. Co-interior angles. Find x .



Answer: _____

15. Angle sum. Find x .



Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Sketch the shape and mark equal sides, parallel sides and right angles.
- Check your four angles add to 360° when you finish.

Read each one, draw your own diagram, then answer the question.

16. A quadrilateral has three right angles. Draw it and explain what the fourth angle must be.

Draw your diagram here

Answer: _____

17. Explain why every square is also a rectangle and also a rhombus. Use sketches to help.

Draw your diagram here

Answer: _____

18. The angles of a quadrilateral are in the ratio $1 : 2 : 3 : 4$. Draw a sketch and find all four angles.

Draw your diagram here

Answer: _____

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · 18 questions

NSW syllabus outcome

MA4-GEO-C-01 — applies the properties of triangles and quadrilaterals to solve problems · MAO-WM-01

NAME _____

PAGE 12 OF 13

This sheet mixes every skill from the Geometrical Figures chapter. Work through it as though it were the topic test.

EASY

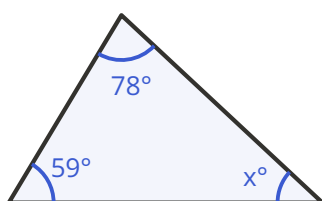
9 QUESTIONS

QUICK REMINDERS

- Triangle angles add to **180°** · quadrilateral angles add to **360°**.
- Ticks show equal sides; an isosceles triangle has two equal base angles.
- An exterior angle equals the two opposite interior angles added together.

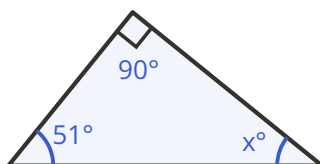
Find the size of the angle marked x.

1. Find x.



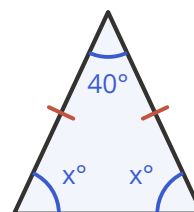
Answer: _____

2. Find x.



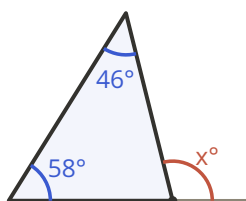
Answer: _____

3. Find x.



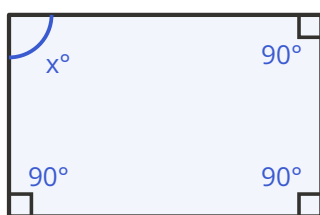
Answer: _____

4. Find x.



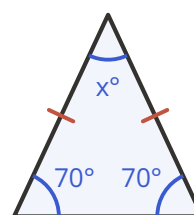
Answer: _____

5. Find x.



Answer: _____

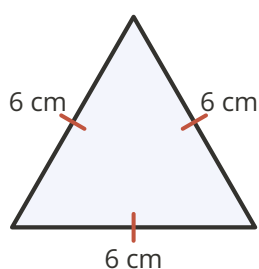
6. Find x.



Answer: _____

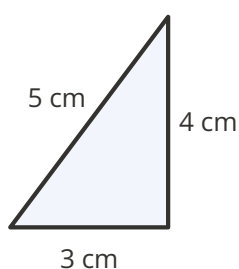
Name each shape.

7. Name by its sides.



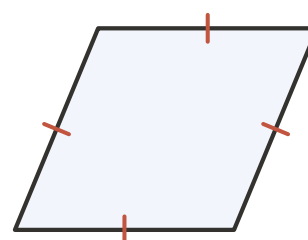
Answer: _____

8. Name by its sides.



Answer: _____

9. Name this shape.



Answer: _____

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · continued

PAGE 13 OF 13

MEDIUM

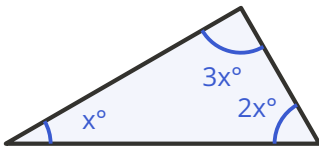
6 QUESTIONS

CHOOSING THE RIGHT RULE

- Count the sides first: 3 sides means 180° , 4 sides means 360° .
- Look for ticks (equal sides) and arrows (parallel lines) before you start.

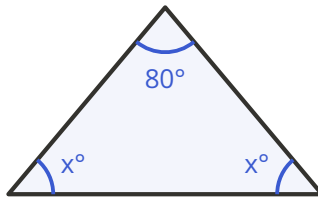
Find the value of x .

10. Find x .



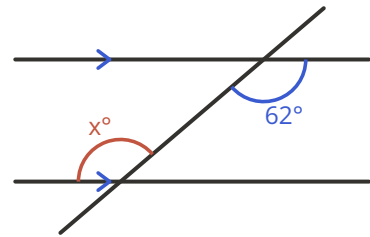
Answer: _____

11. Find x .



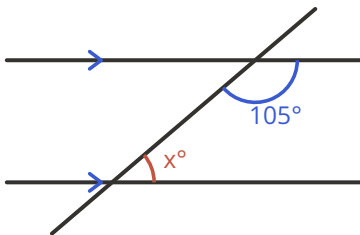
Answer: _____

12. Find x .



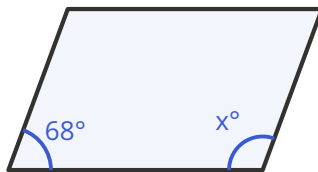
Answer: _____

13. Find x .



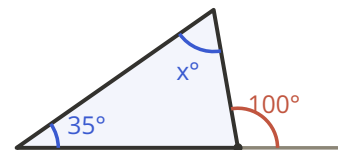
Answer: _____

14. Find x .



Answer: _____

15. Find x .



Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Draw the shape, write on every angle you know, then decide which rule to use.
- Always finish by writing a short reason for your answer.

Read each one, draw your own diagram, then answer the question.

16. Classify a triangle with angles 100° , 40° and 40° by its sides and by its angles. Draw and label it.

Draw your diagram here

Answer: _____

17. A line through the apex of an isosceles triangle is drawn parallel to the base. The base angles are both 50° . Draw the diagram and find the two angles at the apex.

Draw your diagram here

Answer: _____

18. The angles of a triangle are in the ratio $1 : 2 : 3$. Draw a sketch, find all three angles, then classify the triangle.

Draw your diagram here

Answer: _____